BUSINESS ANALYTICS CASE STUDY

Retail Sales Business Analysis

A Business Analytics Case Study Using SQL & Python — Uncovering Sales, Customer, and Profitability Insights for Management Decision-Making

PROJECT NAME Retail Sales Business Analysis	**DOMAIN** Retail & Business Analytics
TOOLS USED SQL Server, Python, Pandas, Matplotlib	**DATASET** Retail Superstore Sales Dataset
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PREPARED BY

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Product Analytics · Business Analytics · AI

INTRODUCTION

About the Author

Jaskaran Arora

PRODUCT ANALYTICS | BUSINESS ANALYTICS | AI

I am a final-year Computer Science student with a strong interest in Product Analytics, Business Analytics, and applied AI. I enjoy working with data to understand how businesses actually make money, where they lose it, and what decisions can move the needle. My approach combines technical tools such as SQL and Python with a business-first mindset, so that every analysis ends in a clear, actionable recommendation rather than just a chart.

I have hands-on experience across product teardowns, business intelligence dashboards, and data-driven case studies like this one. I am currently seeking opportunities in Product Analyst, Business Analyst, and Data/BI roles where I can help teams turn raw data into confident business decisions.

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SECTION 01

Executive Summary

Every retail business generates one thing in large volume: transactions. But transactions alone don't tell management anything useful. A business can be doing thousands of sales a month and still be quietly losing money on specific products, customers, or regions — and no one would know unless the data is properly analyzed.

This is the core problem this project addresses. Looking only at total sales gives a false sense of security. A company can have record-breaking revenue and a shrinking profit margin at the same time, simply because of discounting habits, unprofitable product lines, or over-reliance on a small customer base. Total sales is a headline number — it is not a decision-making tool.

Data analytics changes this. By breaking sales down by time, region, category, customer, and discount level, management gets a much clearer picture of what is actually driving the business — and what is quietly working against it. This is exactly what this project set out to do.

2	5	6
ANALYTICS ENGINES SQL + PYTHON	BUSINESS DIMENSIONS ANALYZED	RECOMMENDATIONS DELIVERED

Using SQL for structured business queries and Python for data cleaning, exploration, and visualization, this project analyzes the Retail Superstore Sales Dataset to answer the questions that matter most to management — which products perform, which customers matter most, which regions need attention, and whether the current discounting strategy is helping or hurting profit.

> ◆ **WHAT THIS PROJECT DELIVERS**
>
> A complete, end-to-end business analysis — from raw data to clean insights to concrete recommendations — covering sales performance, customer behavior, product profitability, regional trends, and the real impact of discounting.

SECTION 02

Business Problem

Imagine a retail company that generates thousands of orders every year, across dozens of product categories, hundreds of customers, and multiple regions. On the surface, the business looks healthy — orders keep coming in. But management needs answers that "total sales" simply cannot give them.

Specifically, leadership wants to know:

- Which products generate the highest revenue?
- Which products are actually losing money?
- Which customers generate the maximum sales?
- Which regions are performing the best?
- Does discounting actually increase profit, or quietly erode it?
- Are sales increasing year over year?
- Is there a seasonal pattern to demand?

Answering these manually — by scrolling through spreadsheets — is close to impossible once the data grows beyond a few hundred rows. Patterns that matter, like a profitable-looking product that actually loses money after discounts, stay hidden in plain sight. This is exactly why SQL and Python were used: SQL for fast, structured business queries, and Python for cleaning, exploring, and visualizing the data.

SECTION 03

Dataset Overview

This project uses the **Retail Superstore Sales Dataset**, a widely used retail transactions dataset that mirrors how a real mid-sized retail business records its orders — every row represents one product sold as part of one order, along with where, to whom, and at what price.

Field	What It Represents	Why It Matters to Management
Order Date	The date the order was placed	Reveals sales trends and seasonality over time
Customer	The buyer of the order	Identifies top customers and buying behavior
Region	Geographic area of the sale	Highlights strong and weak performing markets
Category	High-level product grouping	Shows which product lines drive the business
Sub-Category	Detailed product grouping	Pinpoints specific product performance
Sales	Revenue generated by the order	The starting point for all revenue analysis
Profit	Actual money earned after costs	The real measure of business health, not just sales
Discount	Price reduction applied to the order	Used to test whether discounting helps or hurts profit
Quantity	Units sold in the order	Supports demand and inventory planning
Product Name	The specific item sold	Enables product-level performance ranking

SECTION 04

Methodology

The project followed a structured, six-step workflow — the same approach a Business Analyst would use on a real client engagement.

1

Understanding the Dataset

Reviewed every field in the dataset and mapped each one to a real business question before writing a single line of code.

2

Cleaning the Data

- Checked for missing values across all key fields
- Identified and removed duplicate records
- Removed columns that added no business value
- Converted date fields into a usable time format

3

Exploratory Data Analysis (EDA)

Calculated core business KPIs and explored sales trends, customer trends, category performance, regional performance, and profitability patterns.

4

Business Visualization

Built line charts (trends over time), bar charts (comparisons across categories, regions and products) and scatter plots (discount vs. profit relationship).

5

SQL Business Analysis

Ran structured business queries using aggregations, GROUP BY and ORDER BY logic, and window functions to rank top products and top customers.

6

Business Recommendations

Converted every insight into a specific, actionable recommendation that management could act on immediately.

SECTION 05

Key Findings

The analysis surfaced eight findings that materially change how management should think about the business.

Sales steadily increased over time

Year-over-year revenue shows a consistent upward trend, indicating healthy underlying demand rather than a one-off spike.

Q4 generates the highest revenue

The final quarter of the year consistently outperforms the rest, pointing to a clear seasonal buying pattern.

Technology category performs best

Technology products generate the strongest combination of sales volume and profit in the business.

Some high-revenue products generate low profit

Several top-selling products contribute very little profit once costs and discounts are factored in.

Discounts often reduce profitability

Higher discount levels are frequently associated with lower — sometimes negative — profit margins.

A small % of customers drive most revenue

A small group of customers accounts for a disproportionately large share of total sales.

Regional sales vary significantly

Some regions consistently outperform others, showing room for targeted improvement.

Standard shipping is most commonly used

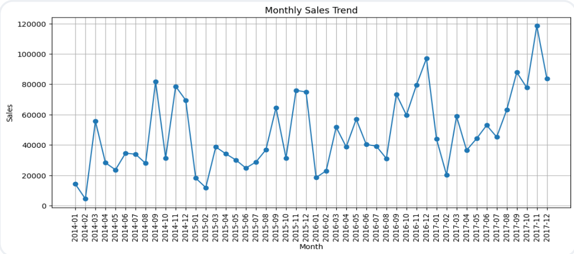
Most customers default to standard shipping, which has implications for logistics planning.

SECTION 06

Visual Analysis

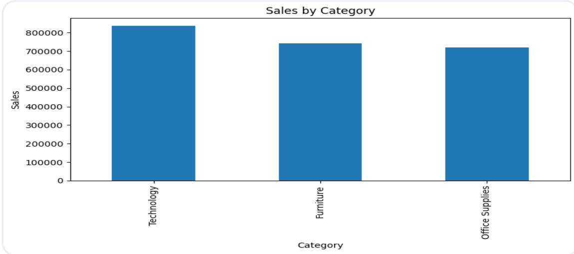
Charts turn raw numbers into patterns a manager can spot in seconds. Each visualization below was chosen for a specific business reason.

Monthly Sales Trend



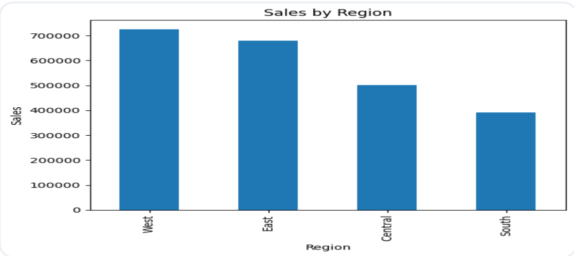
Why: Shows how revenue moves over time. **Insight:** Steady growth with a clear Q4 spike. **Use:** Plan cash flow and staffing around predictable high/low months.

Sales by Category



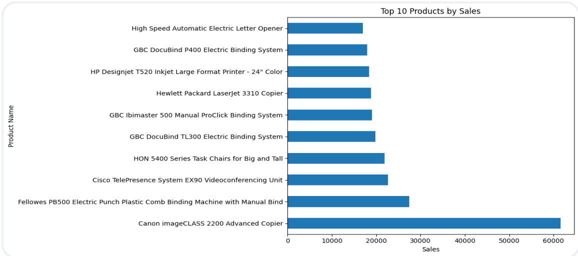
Why: Compares category performance side by side. **Insight:** Technology leads, Furniture lags. **Use:** Prioritize budget toward categories that perform.

Sales by Region



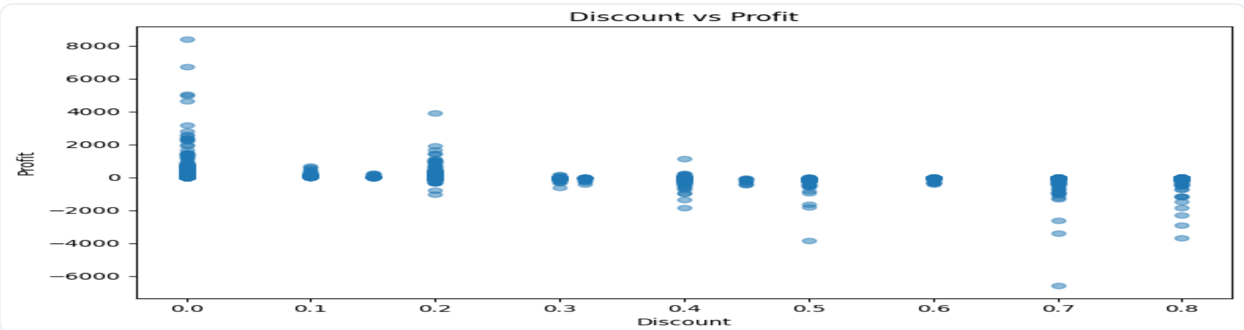
Why: Identifies geographic strengths and weaknesses. **Insight:** Performance is uneven across regions. **Use:** Direct resources toward markets that need support.

Top Products



Why: Ranks the business's best-selling products. **Insight:** A small set of products carries most revenue. **Use:** Protect stock for these core products.

Discount vs. Profit



Why: Tests whether discounting actually helps the business. **Insight:** Profit tends to fall — and can turn negative — as discount level rises. **Use:** Set clear limits on how much discount can be offered before a sale stops being worthwhile.

SECTION 07

SQL Analysis

SQL was used as the primary engine for structured business queries — the exact type of questions a manager would ask in a Monday morning review. The table below summarizes the business purpose behind each analysis performed.

Business Query	Business Purpose
Total Sales & Total Profit	Establish the overall size and health of the business
Top Customers	Identify who to prioritize for retention and loyalty programs
Top Products	Identify which products deserve more shelf space and marketing focus
Regional Performance	Compare markets to identify where to invest or intervene
Category & Sub-Category Performance	Understand which product lines actually drive the business
Year-Wise Sales	Confirm growth trends and support forecasting
Ranking Products & Customers (Window Functions)	Create clean, ranked leaderboards for management reporting

◆ HOW SQL WAS USED

Every query was built around aggregations (summing sales and profit), GROUP BY logic (breaking totals down by category, region, or customer), and ORDER BY logic (ranking the results from highest to lowest). Window functions were then used to rank products and customers cleanly, without needing separate queries for each ranking. This kept the analysis fast, repeatable, and easy to hand off to any team that needs updated numbers in the future.

SECTION 08

Business Recommendations

Every insight in this project was translated into a specific action management can take.

01 **Reduce Unnecessary Discounts**

Cap discounts at levels shown to still protect profit margin, rather than approving them case by case.

02 **Invest More in Technology Products**

Direct more marketing budget and inventory toward the category already proven to deliver strong returns.

03 **Focus on High-Value Customers**

Build loyalty and retention programs around the small group of customers who generate a large share of revenue.

04 **Improve Underperforming Regions**

Investigate why certain regions lag behind and pilot targeted promotions or staffing changes there.

05 **Plan Inventory for Seasonal Demand**

Increase stock and staffing ahead of Q4, given its consistently higher demand.

06 **Segment Customers, Refine Pricing**

Group customers by value and behavior to tailor pricing and offers, instead of one blanket policy.

SECTION 09

Business Impact

This analysis is not just a reporting exercise — it directly supports how management runs the business.

Better Decision Making

Replaces guesswork with evidence when deciding where to invest time, budget, and attention.

Improved Profitability

Highlights exactly where discounting and low-margin products quietly eat into profit.

Smarter Inventory Planning

Seasonal and product-level trends help avoid both stockouts and overstocking.

Deeper Customer Understanding

Identifies exactly which customers matter most, enabling focused retention efforts.

Stronger Forecasting

Historical trend patterns provide a reliable foundation for predicting future sales and planning ahead with confidence.

SECTION 10

Skills Demonstrated

- Business Analytics
- SQL
- Python
- Pandas
- Data Cleaning
- Exploratory Data Analysis
- Business Intelligence
- Data Visualization
- Business Storytelling
- Analytical Thinking

SECTION 11

Conclusion

This project set out to answer a simple but important question: what is really happening beneath the top-line sales number of a retail business?

By combining SQL for structured business queries with Python for cleaning, exploration, and visualization, raw transactional data was transformed into a clear, decision-ready picture — covering sales performance, customer value, product profitability, regional trends, and the true impact of discounting.

The biggest takeaway is simple: **analytics is not about creating charts.** It is about giving management the confidence to make better decisions — where to invest, where to cut back, and where to pay closer attention. That is the real value this project delivers.

END OF CASE STUDY

Thank You for Reading

I hope this case study reflects how I think about business problems and turn data into decisions. I'd love to connect for feedback, questions, or collaboration opportunities.

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